

MODULE 2 OF 6

Sampling, Data and Database Integrity

An estimate can never be better than the samples it is built from

This module covers the part of the value chain that quietly determines the accuracy of everything downstream — and the part most often done carelessly. We derive the theory of sampling error, then apply it to the real integrity problems sitting inside the Toro database now.

LEARNING OBJECTIVES

- Quantify the fundamental sampling error using Gy's theory and explain why particle size dominates it.
- Design appropriate sampling for an alluvial placer and for a primary hard-rock target.
- Structure a drillhole / pit database into its four core tables.
- Determine bulk density correctly and quantify its effect on declared tonnage.
- Build a QAQC system from standards, blanks and duplicates.
- Audit the Toro dataset and composite it correctly.

1 The error budget

A Toro pit bulk sample is roughly 100–200 kg; a future mining operation will process several million tonnes. Every tonne mined will have its grade *predicted* from a sampled fraction of the order of one part in ten million. The total error in a final block estimate is a sum of variances from independent stages:

$$\sigma_{\text{total}}^2 = \sigma_{\text{sampling}}^2 + \sigma_{\text{sub-sampling}}^2 + \sigma_{\text{assay}}^2 + \sigma_{\text{estimation}}^2 \quad (2.1)$$

The order that matters

Modules 4 and 5 reduce only $\sigma_{\text{estimation}}^2$. If the sampling and assay variances are large, no kriging plan can rescue the estimate — you would be interpolating noise with great precision. Sampling quality is logically prior to geostatistics.

2 Gy's theory of sampling

Pierre Gy gave the industry a quantitative theory of sampling error. Its centrepiece is the **Fundamental Sampling Error** (FSE): the irreducible error that remains even with a perfect protocol, arising because ore is made of discrete, heterogeneous particles. *Constitution heterogeneity* is intrinsic and cannot be mixed away; *distribution heterogeneity* is the spatial arrangement and can be reduced by homogenisation. The FSE flows from constitution heterogeneity. For a low-grade ore sampled correctly:

$$\sigma_{\text{FSE}}^2 = \left(\frac{1}{M_S} - \frac{1}{M_L} \right) f g c \ell d^3 \quad (2.2)$$

Here M_S is the sample mass, M_L the lot mass, d the nominal top particle size (note the *cube*), f the shape factor (≈ 0.5), g the granulometric factor (≈ 0.25), c the mineralogical factor and ℓ the liberation factor. The mineralogical factor carries the grade dependence:

$$c = \frac{1 - a_L}{a_L} [(1 - a_L)\rho_M + a_L\rho_g] \xrightarrow{a_L \rightarrow 0} \frac{\rho_M}{a_L} \quad (2.3)$$

Two consequences must shape how you sample. **Particle size dominates, because of the cube**: halving the top size by crushing cuts the sampling variance to one eighth — which is why a sampling protocol is a staged sequence of crush, split, crush, split. **Low grade plus a dense, coarse mineral is the worst case**: since $c \propto 1/a_L$, error explodes as grade falls. Cassiterite is dense ($\rho \approx 6.9 \text{ g/cm}^3$), often coarse and frequently liberated — it behaves like a coarse-gold ore, where a few heavy grains carry a disproportionate share of the metal.

The FSE is a variance that exists even between two samples taken at the *same* place. On a variogram (Module 4) that same-location variance appears as the intercept at zero distance — the **nugget effect**. A large part of a tin deposit's nugget is not geology; it is Gy's fundamental sampling error plus assay error. Improving the sampling protocol literally lowers the nugget you fight in Module 4.

Beyond the FSE, Gy catalogued *avoidable* errors: the grouping and segregation error (controlled by taking many small increments) and the materialisation errors of delimitation, extraction and preparation. These produce *bias*, which is far more dangerous than random error because no amount of repeat sampling reveals it.

3 Sampling the two Toro orebodies

The two Toro targets are sampled by different methods because they are different estimation problems.

3.1 The alluvial target: a volume-grade problem

Alluvial cassiterite sits in wash gravels beneath barren overburden. The economic quantity is a concentration of heavy mineral per unit volume, reported in **grams of tin per cubic metre**. A pit is dug to the wash layer, a measured bulk sample of gravel is washed and panned to a heavy-mineral concentrate (HMC), the HMC is weighed and assayed. The in-situ grade is a *chain* of factors:

$$g_{\text{Sn}} \left[\frac{\text{g}}{\text{m}^3} \right] = \underbrace{\frac{M_{\text{HMC}}}{M_{\text{bulk}}}}_{\text{yield}} \times a_{\text{Sn}} \times \rho_{\text{gravel}} \times 10^6 \quad (2.4)$$

The Toro XRF dataset reports the grade *of the panned heavy-mineral concentrate*, not the grade of the ground. A sample reading “3.97% SnO₂” describes a kilogram of concentrate, not a cubic metre of gravel. To turn it into a resource grade you must multiply by the HMC yield and the bulk density, as in equation (2.4). Treating concentrate assays as in-situ grades would overstate the deposit by one to two orders of magnitude — the classic, fatal error in alluvial tin reporting.

The chemistry confirms it: raw alluvium would assay 70–90% SiO₂, but the Toro field samples average roughly 32% SiO₂ and 29% Fe₂O₃ — these are heavy-mineral concentrates.

3.2 The primary target: a grade-tonnage problem

The primary greisen and pegmatite target is sampled like any hard-rock deposit, in % **Sn** of rock in place: diamond core drilling (the definitive method), reverse-circulation drilling, channel sampling of veins, and pitting of weathered cover. Two cautions: **true width** (grade applies to a thickness perpendicular to the structure, not the down-hole interval) and **refractory chemistry** (cassiterite resists aqua-regia digestion, so analyses must specify four-acid digestion or lithium-borate fusion XRF — which the ANRML26-137 laboratory correctly used).

4 The drillhole and pit database

Estimation software consumes a relational database of four core tables, linked by a unique hole or pit identifier: **Collar** (location and depth), **Survey** (the hole path), **Assay** (grade per interval), and **Lithology** (the geological log that defines domains). Every interval has a from/to depth; intervals never overlap or gap; coordinates use one declared system (for Toro, WGS84 UTM Zone 32N). The Toro Priority Drilling Points file is already a well-formed collar table in embryo — 13 collars at Bishiwai, Yadabongo and Kadade.

5 Bulk density — the silent half of every tonne

Tonnage is volume times density, and a proportional error in density is a proportional error in tonnage and contained metal, one-for-one:

$$T = V \times \rho \quad \implies \quad \frac{\Delta T}{T} = \frac{\Delta \rho}{\rho} \quad (2.5)$$

Density must be measured by domain and weathering state, not assumed. For the unconsolidated Toro wash gravel, core displacement is impossible: an in-situ method (a measured excavation, weighed, with a surveyed void volume) is required, recording the natural moisture state. For the primary granite, fresh rock is near 2.6–2.7 t/m³ but the weathered cupola hosting near-surface tin can be far lower.

6 QAQC: deciding whether the database can be trusted

QAQC separates two properties of the data. **Accuracy** is freedom from bias; **precision** is repeatability. They are independent, and demonstrated by three control instruments: **Certified Reference Materials** test accuracy, **blanks** test contamination, and **duplicates** test precision — each inserted blind at roughly one in twenty. Precision is read from duplicate pairs as the relative difference:

$$RD = \frac{|x_1 - x_2|}{\frac{1}{2}(x_1 + x_2)} \times 100\% \quad (2.6)$$

The Toro file contains paired records at identical coordinates — for example DBP-033A and DBP-033A', reading 7.59% and 2.28% SnO₂, a relative difference above 100%. But you cannot tell *why*: the suffixes might mean two stratigraphic layers (real geology) or a re-split (poor precision). The reconnaissance sampling had no QAQC structure, so the explanations are confounded. That is why a designed QAQC protocol — blind, randomly inserted duplicates — must run through the next campaign from the first sample.

7 Worked example: auditing the Toro database

The file `Motimose_GeoChemData.csv` holds 40 XRF records. Some are not in-situ samples at all: identifiers carrying mineral-separation tags (Nb, Fe, Zr, SnO₂+SiO₂) are laboratory heavy-mineral fractions. The record DBP-010 SnO₂+SiO₂-2 assays 65.4% SnO₂ — a lab-made cassiterite concentrate that exists nowhere in the ground. It must be filtered out.

```
import pandas as pd, numpy as np
geo = pd.read_csv("Motimose_GeoChemData.csv")

# IDs carrying a mineral-separation tag are laboratory fractions,
# NOT in-situ field samples - they must not enter grade statistics.
sep_tags = ["Nb", "Fe", "Zr", "SnO2+SiO2", "FeB", "SnO2 (2)"]
def is_separation(sid):
    s = str(sid)
    return any(t in s for t in sep_tags) and len(s.split()) > 1

geo["record_type"] = np.where(
    geo["Lab_Sample_ID"].apply(is_separation),
    "lab_separation", "field_sample")
# field_sample 32 ; lab_separation 8
```

Eight of 40 records are laboratory fractions; 32 are genuine field samples. Those 32 still over-represent pits logged with two layers, so we **composite** to one grade per location:

```
fs = geo[geo.record_type == "field_sample"].copy()
comp = (fs.groupby("Field_Pit_ID")
        .agg(SnO2=("SnO2", "mean")) # length-weight if logged
        .reset_index())
# 32 field samples -> 26 pit locations
```

The audit result

The raw “40-sample” dataset resolves into **26 pit locations** with a defensible in-situ concentrate grade. That number — not 40 — is the real spatial dataset for Modules 3–5. Proper compositing should be length-weighted; recording sampled thickness on every future pit is a prerequisite for a correct composite.

Module 2 of the Toro Resource Estimation course. Next: Module 3, Exploratory Data Analysis and the Statistics of Grade.